Abstract

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We specify a reproducible advanced literature-review workflow for rapidly changing domains in which a single query does not adequately represent methodological or temporal variation. The exemplar is configured for exoplanet atmospheric composition and separates foundation, James Webb Space Telescope, and molecular-detection phases.

The design is informed by systematic-review reporting practice (Page et al. 2021), but this repository release is a methods exemplar rather than a completed systematic review: its default corpus is synthetic and its phase boundaries, search coverage, and domain interpretation require review for any live application.

The pipeline combines multi-engine retrieval, canonical de-duplication, deterministic screening, optional LLM-assisted classification, full-text assessment, bibliometrics, knowledge-graph construction, visualization, reproducibility assessment, and export. Each stage has a declared input, output, method reference, and definition of done in `methods_pipeline.yaml`. The numbered scripts are thin adapters; reusable computation is tested in `src/`.

The manuscript is generated from configuration and recorded artifacts. Offline runs use a clearly labelled synthetic fixture corpus with reserved identifiers and generated authors; they exercise pipeline behavior and are not evidence about exoplanet atmospheres. Live runs must retain retrieval reports, source provenance, claim classifications, and the exact configuration needed to interpret reported results.

The contribution is therefore infrastructural: it makes phase-aware search design, evidence boundaries, accessibility metadata, and reproducibility checks explicit while leaving domain claims subject to source-backed review.

Keywords: exoplanet atmospheres, literature review pipeline, systematic-review reporting, multi-phase search, LLM filtering, James Webb Space Telescope, atmospheric composition, transit spectroscopy, bibliometrics, cross-validation